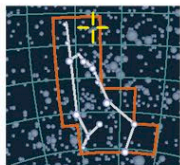


PLANETARY NEBULA

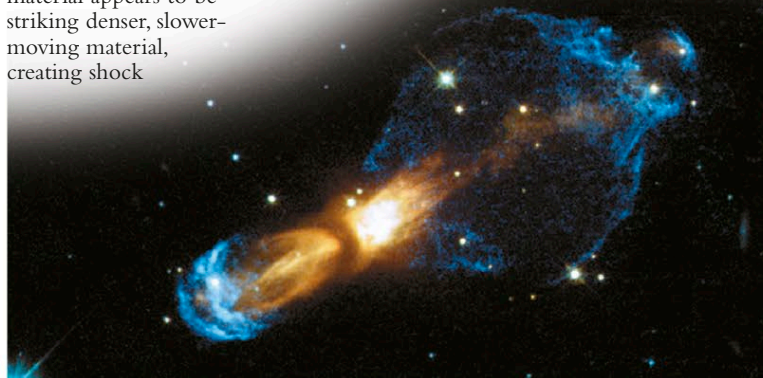
Calabash Nebula



CATALOG NUMBER
OH231.8+4.2
DISTANCE FROM SUN
5,000 light-years
MAGNITUDE 9.47

PUPPIS

One of the most dynamic planetary nebulae, the Calabash Nebula's central star is expelling gas at a speed of 435,000 mph (700,000 kph). The fast-moving material is being channeled into streamers on one side and into a jet on the other. The jet of material appears to be striking denser, slower-moving material, creating shock



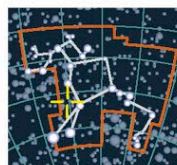
waves. Radio observations have revealed an unusually large amount of sulfur in the gas around the star, which may have been produced by the shock waves. This planetary nebula is in the earliest stages of formation and has offered astronomers the chance to observe the kind of processes that led to the creation of more established planetary nebulae elsewhere in the Milky Way.

ROTTEN EGG NEBULA

The Calabash Nebula is popularly called the Rotten Egg Nebula because it contains a lot of sulfur, which smells like rotten eggs. The outflows of expelled gas show up bright yellow-orange in the center of this picture.

PLANETARY NEBULA

Southern Crab Nebula



CATALOG NUMBER
Hen 2-104
DISTANCE FROM SUN
10,000 light-years
MAGNITUDE 14.20

CENTAURUS

This beautiful nebula has a complex "symbiotic" binary star system at its center—a Mira-type pulsating red giant and a white dwarf orbiting so closely that the dwarf's gravity is pulling away gas from the giant's upper atmosphere. The result is a nebula that looks relatively similar to other double-lobed planetaries but is formed by a somewhat different mechanism. A disk of hot material spiraling onto the white dwarf acts as a constraint around the middle of the nebula, while a relatively slow stellar wind of material escaping from the red giant interacts with the much faster wind from the white dwarf to drive the expanding lobes of gas above and below. Although these lobes are expanding in size, the inner and outer structures that can currently be seen both have a similar age.

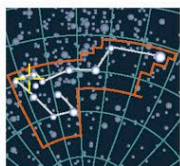


LOBES AND JETS

The Southern Crab's outer lobes glow in cometlike streaks where they plow into surrounding interstellar material. Similarly, narrow jets escaping from above and below the white dwarf's poles only becomes visible where they encounter this material.

BLUE SUPERGIANT

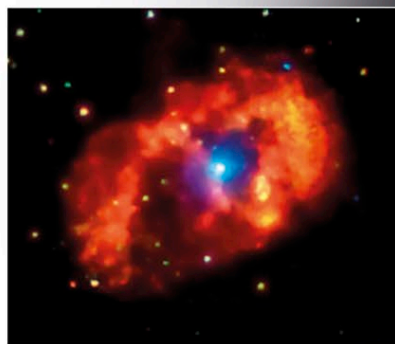
Eta Carinae



DISTANCE FROM SUN
8,000 light-years
MAGNITUDE 6
SPECTRAL TYPE B0

CARINA

Long thought to be a single massive star, Eta Carinae is in fact a binary system of two huge stars, one of which has about 120 times the mass of the Sun and shines more than 5 million times as bright. A vast dumbbell of ejected stellar material, which is expanding at a rate of about 1.2 million mph (2 million kph), hides the stars from direct view. Eta Carinae experiences two types of irregular eruptions—gradual brightening of one to two magnitudes (see pp.232–233) that can last for a few years, and shorter-lived intense outbursts that are accompanied by the ejection of more than a solar mass of material. As a result, Eta Carinae has varied in brightness from as faint as magnitude 7 to as bright as -1 since it was first cataloged by the English astronomer Edmond Halley in 1677. The last major eruption, in 1841, saw its brightness rival Sirius and released the two lobes of material that now surround the system. Eta Carinae survived that outburst and currently hovers around sixth magnitude, but it will probably eventually destroy itself in a supernova.



HOMUNCULUS NEBULA

A Chandra X-ray image reveals super-hot gas created where high-energy stellar winds from the central stars energize the expanding gas cloud, known as the Homunculus.

EXPLODING DUMBBELL

This false-color image shows the dumbbell-shaped clouds of dust and gas first observed being ejected by Eta Carinae about 160 years ago. This is the most luminous star known in the Milky Way, and it could explode in a supernova at any time.

